

SHARON EKULA

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SUMMARY

Recent MS Computer Science graduate from NJIT and AWS Certified AI Practitioner with hands-on experience in applied machine learning, including training and evaluating neural networks, building data pipelines, and benchmarking model performance. Skilled in Python, with practical experience building and evaluating LLM and RAG-based systems, and production experience with Docker and cloud infrastructure. Currently expanding this work into agentic AI to improve system performance and handle more complex tasks. Brings strong analytical thinking, rapid learning ability, and a collaborative approach to applied AI/ML work.

SKILLS

Languages: Python, Java, SQL, JavaScript, TypeScript, HTML, CSS

AI Product Engineering: LLM Integration, Embeddings, RAG Pipeline Design, Prompt Engineering, Evaluation Frameworks, Multi-Turn Agent Memory, Ollama (LLMs)

Machine Learning: TensorFlow, Scikit-learn, Keras, Neural Networks, Model Evaluation

Frontend: React.js, HTML, CSS

Backend & Databases: FastAPI, Spring Boot, REST API Design, PostgreSQL, MySQL

Cloud & DevOps: AWS (EC2, S3, Lambda, IAM, CloudWatch), Docker, Git, GitHub Actions CI/CD

Tools: VS Code, Postman, Linux/Bash

EXPERIENCE

Cisco — Cyber Security Intern

May 2021 – Jul 2021

- Completed Cisco's Cybersecurity Essentials program, gaining foundational knowledge in networking security, cryptography, firewalls, and threat mitigation.
- Designed and simulated a secure university network architecture using Cisco Packet Tracer, implementing layered security, VLANs, and access control policies.

PROJECTS

Gods Word — AI-Powered Product Feature Built End-to-End

Python, ChromaDB, Ollama (Llama 3), Streamlit, YouTube Data API v3

- Built a RAG-based chatbot from scratch (no LangChain), taking it from prototype to a working feature: designing the retrieval logic, ChromaDB vector store, and prompt construction myself, then integrating a locally-hosted LLM (Llama 3) to generate grounded responses from 31,100+ source documents across three data sources.
- Built the evaluation loop myself: checking generated responses against retrieved passages for accuracy and relevance, then iterating on retrieval and chunking logic whenever outputs drifted from the source material.
- Built a full data ingestion pipeline to fetch and process source content, and shipped a production Streamlit application from raw data to a complete, deployed product.
- Currently extending the system with agentic AI capabilities, enabling it to plan multi-step actions and call tools dynamically rather than answering single queries in isolation.

Cloud-Native Image Recognition Service — Computer Vision on AWS

Java, Spring Boot, AWS (EC2, S3, SQS, Lambda, IAM, CloudWatch, Auto Scaling), Docker, Terraform, GitHub Actions

- Deployed a pre-trained CNN via ONNX Runtime as a production image classification service behind a Spring Boot REST API, integrating a deep learning model into a real, callable microservice rather than a standalone script.
- Built a distributed, event-driven pipeline (S3 → SQS → EC2 workers) with Auto Scaling and CloudWatch-based monitoring for reliability under load, with infrastructure as code in Terraform and automated CI/CD via GitHub Actions.

EDUCATION & CERTIFICATIONS

MS, Computer Science — New Jersey Institute of Technology, Newark, NJ (Dec 2025)

B.Tech, Computer Science — Karunya Institute of Technology & Sciences (2023)

Certifications: AWS Certified AI Practitioner, Neural Networks and Deep Learning (Coursera), Complete AI Engineer Training (Udemy: Python, NLP, Transformers, LLMs, LangChain, Hugging Face, APIs)

PUBLICATION

"Hybrid Ensemble Deep Learning for Sleep Quality Prediction Using Wearable IoT Sensors" — ICICT, 2023